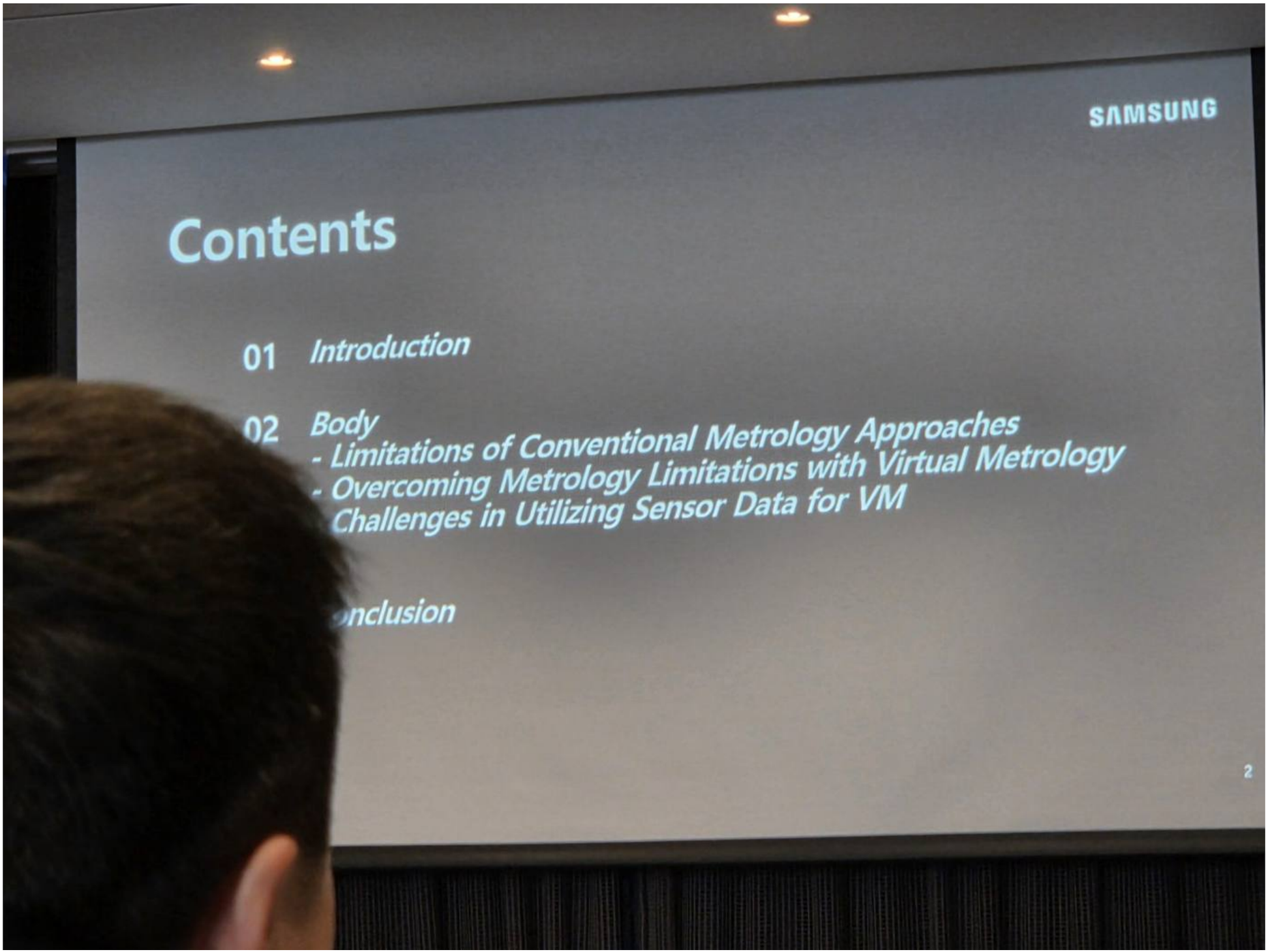
T03 첨단 패키징 가상계측과 이상관리

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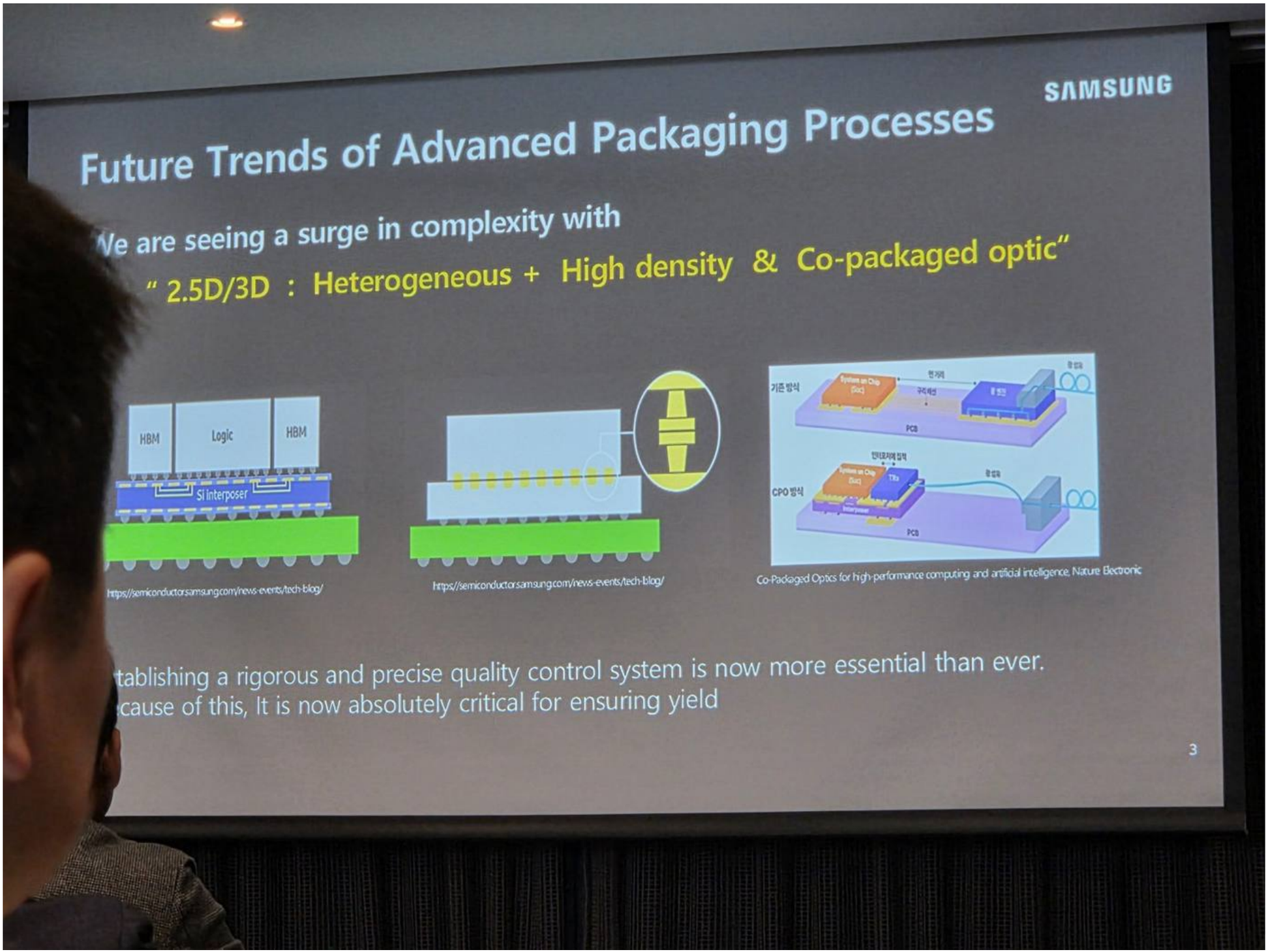
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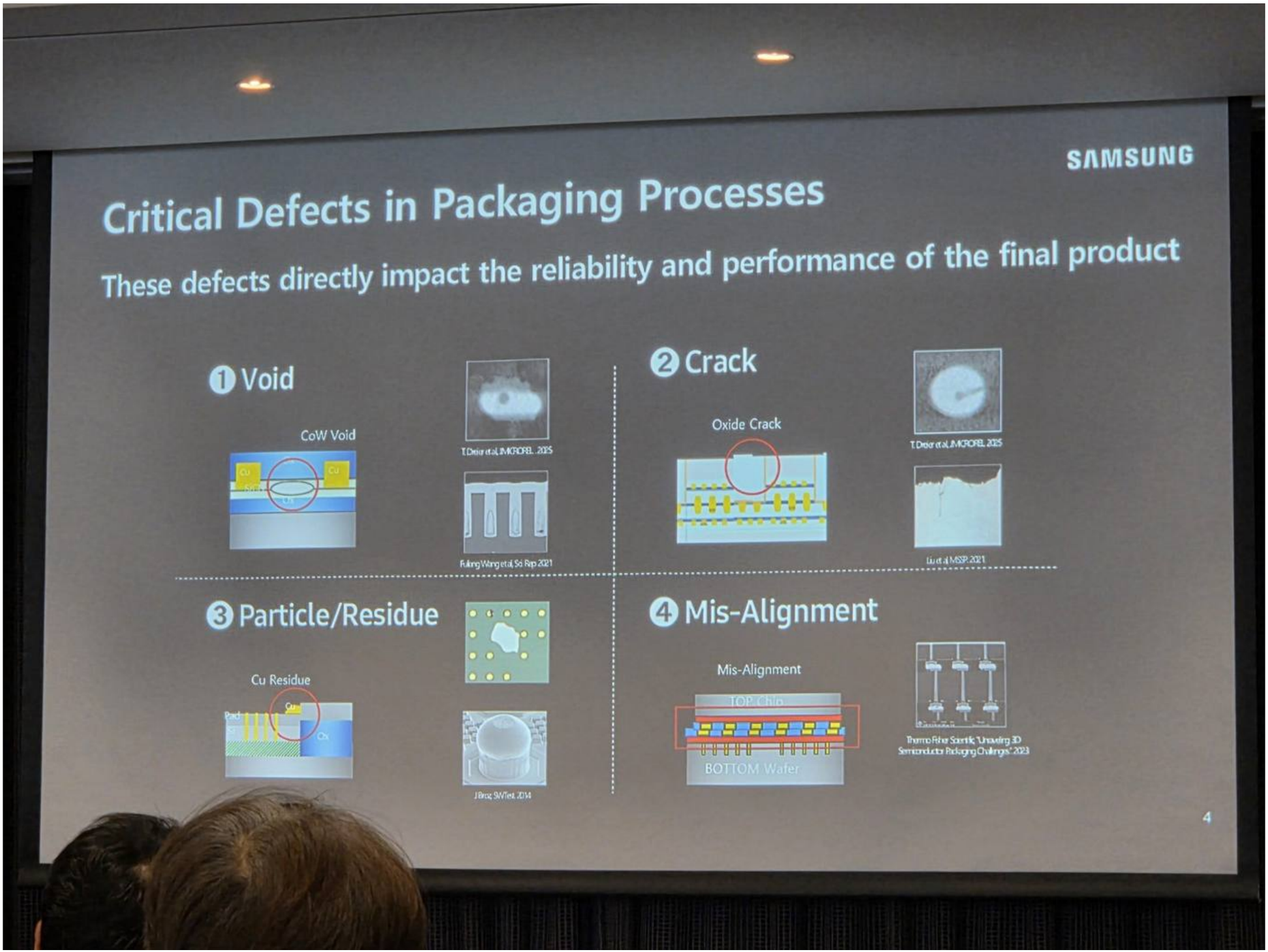
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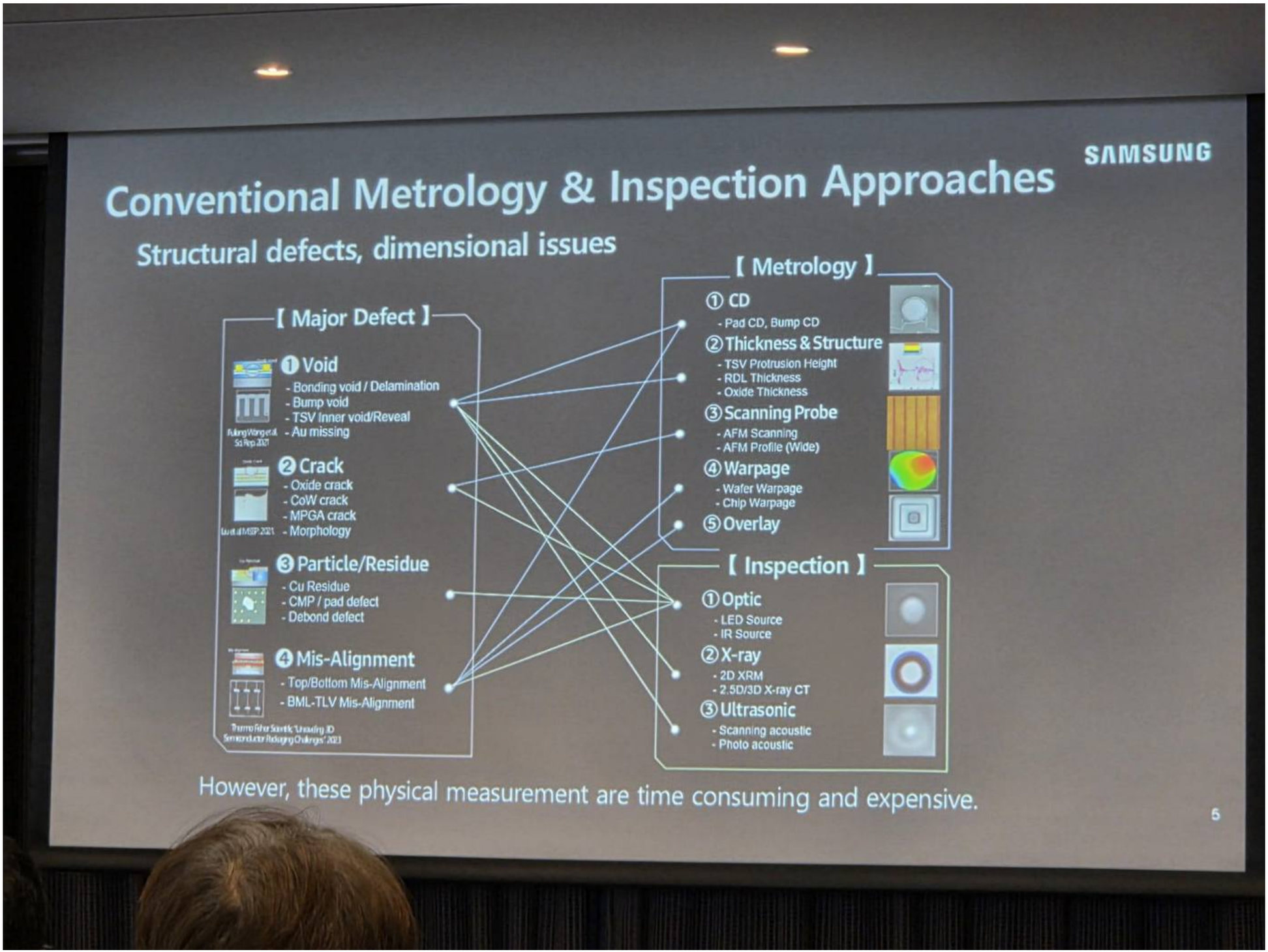
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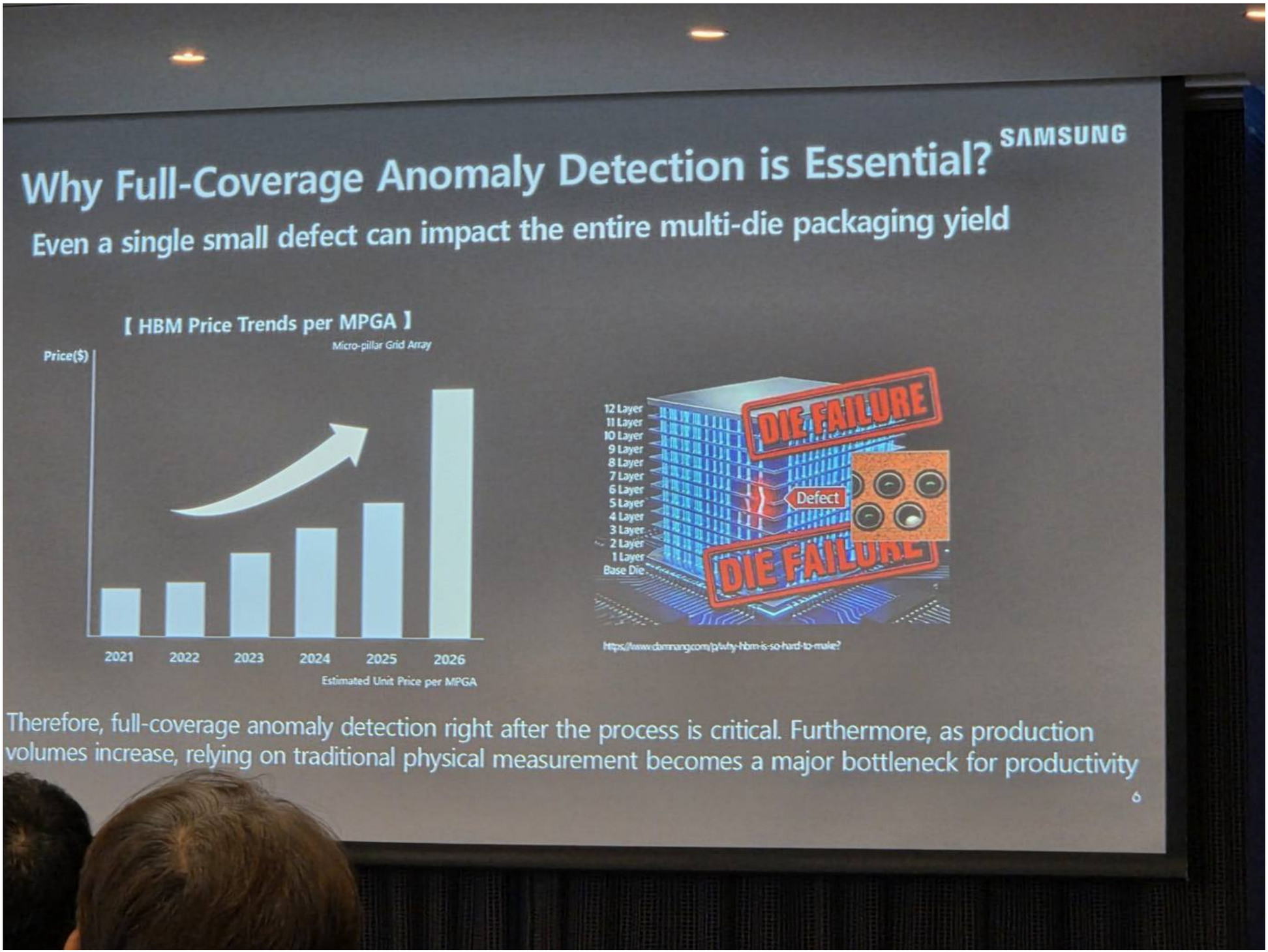
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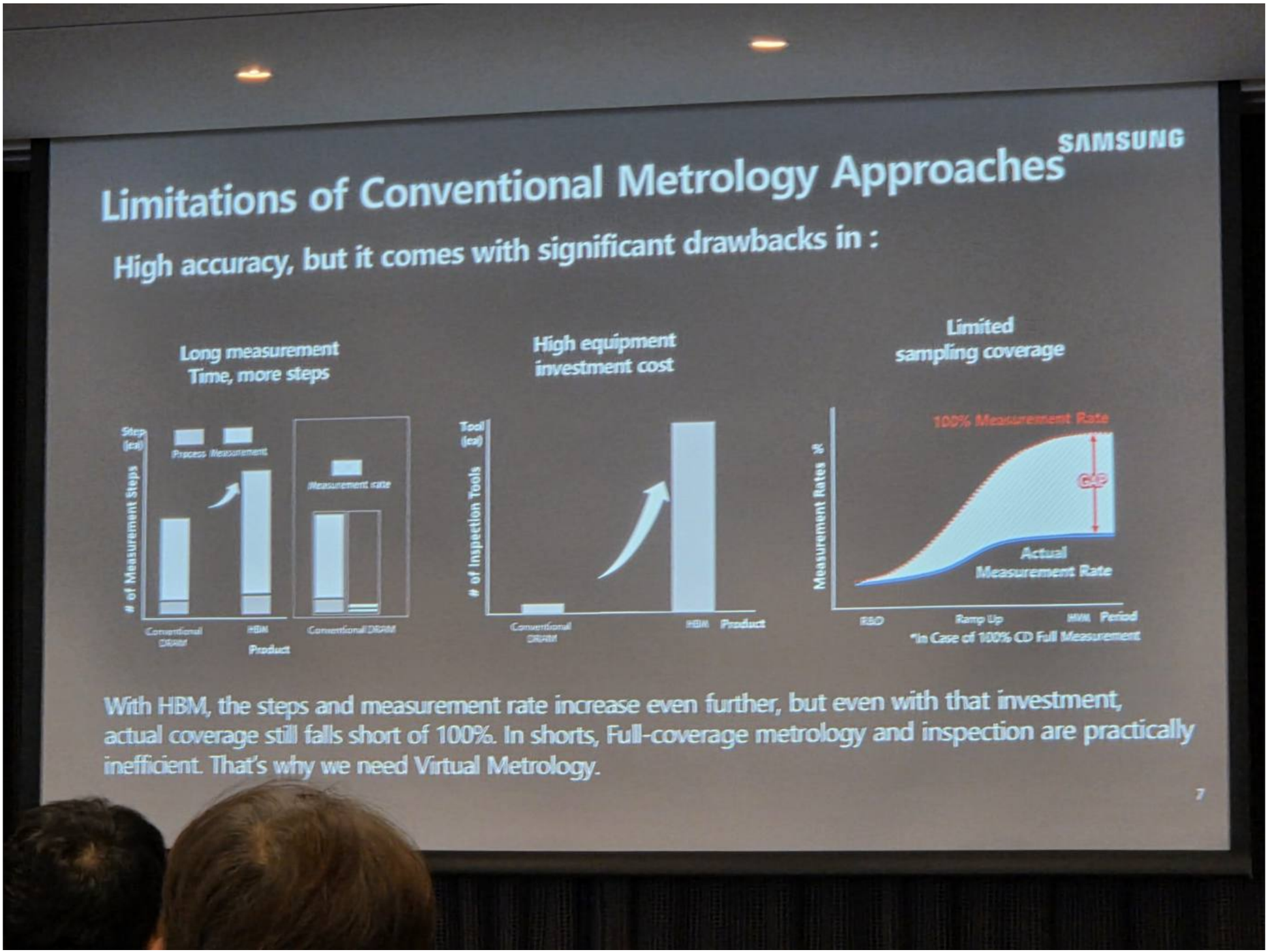
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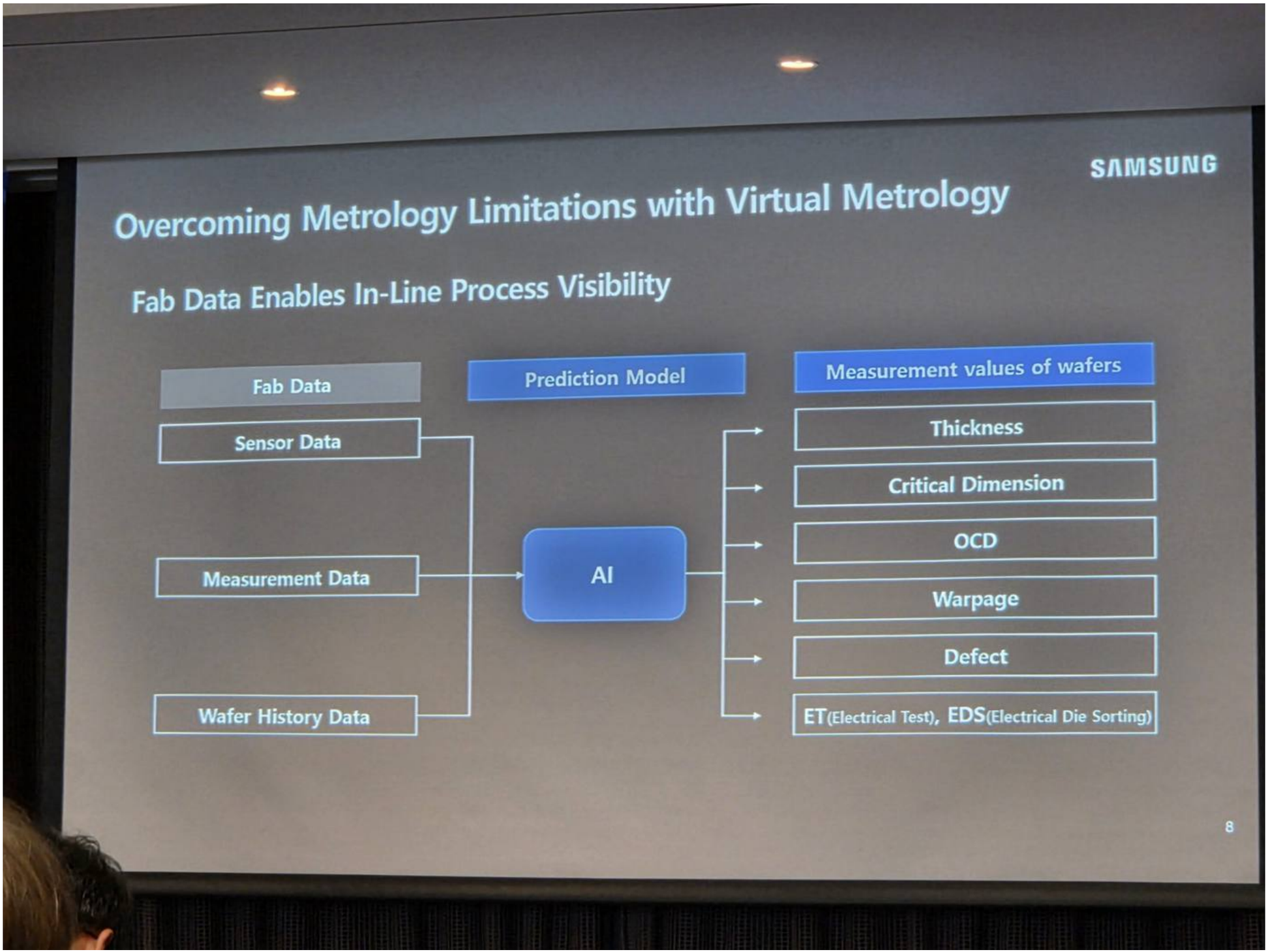
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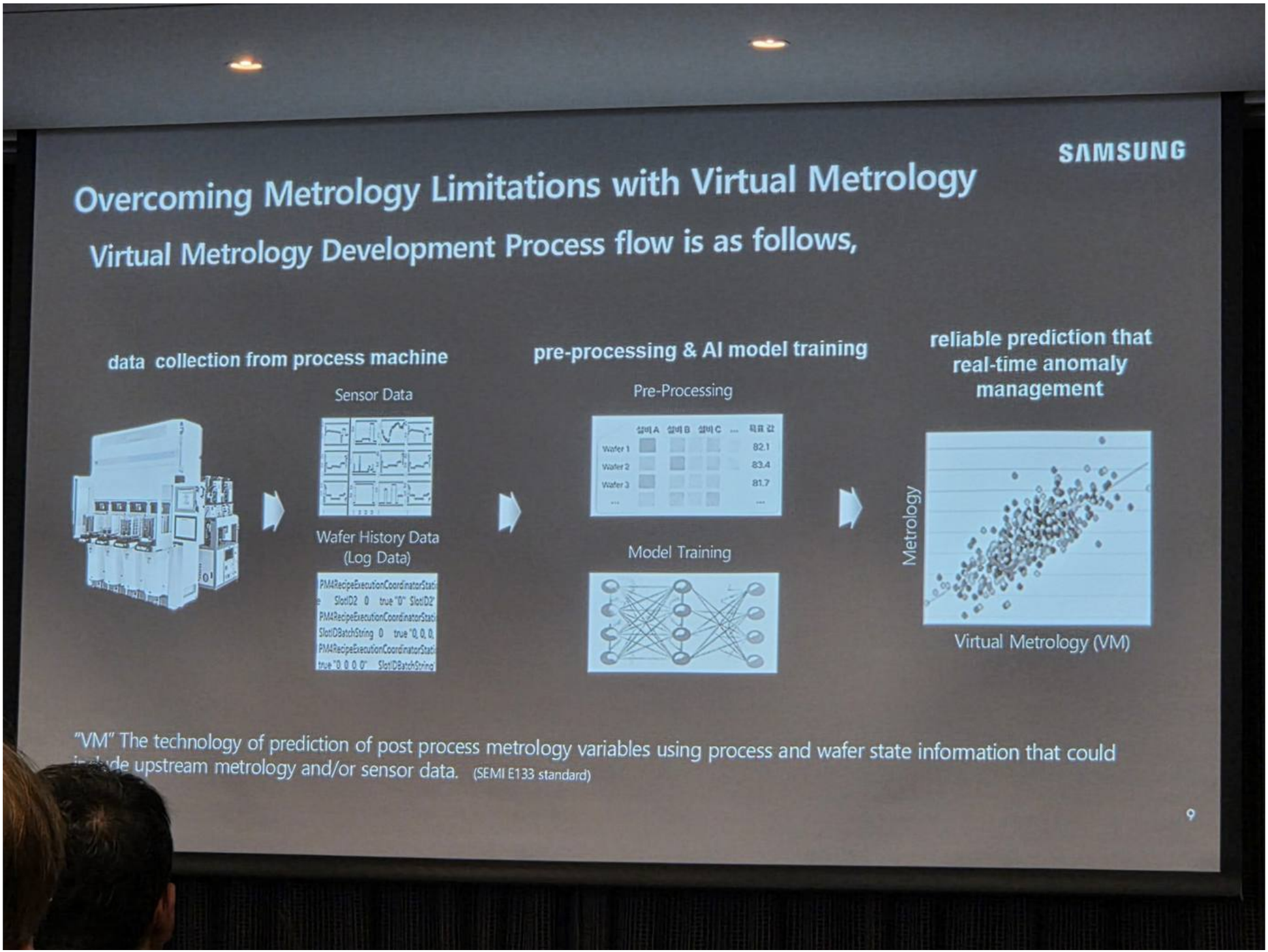
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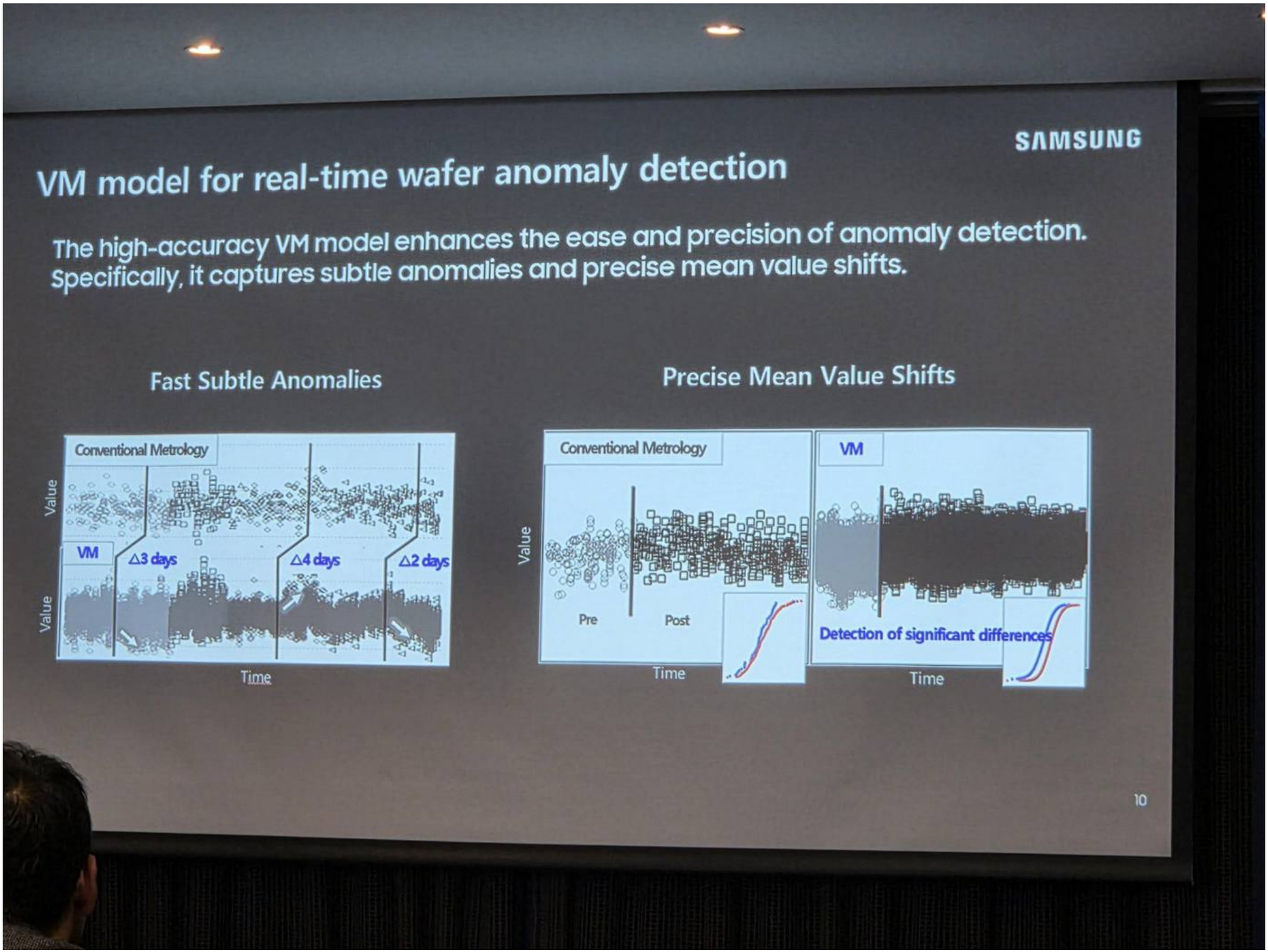


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When does VM provide better anomaly detection than physical sampling measurement?

- When the R^2 for VM consistency exceeds 0.3, the model demonstrates superior performance in detecting mean value shifts compared to 5% sampling measurement.

For VM anomaly detection to be effective,
We need a minimum level of consistency

1	WF level measurement ratio of Metrology									
VM R^2	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90
0.10	1.00	0.95	0.86	0.77	0.69	0.61	0.53	0.44	0.35	0.24
0.20	1.06	1.01	0.91	0.82	0.73	0.65	0.56	0.47	0.37	0.26
0.30	1.14	1.08	0.98	0.88	0.78	0.69	0.60	0.50	0.40	0.27
0.40	1.23	1.17	1.05	0.95	0.85	0.75	0.65	0.54	0.43	0.30
0.50	1.35	1.28	1.15	1.04	0.93	0.82	0.71	0.59	0.47	0.32
0.60	1.50	1.43	1.29	1.16	1.04	0.91	0.79	0.66	0.53	0.36
0.70	1.74	1.65	1.49	1.34	1.20	1.05	0.91	0.77	0.61	0.42
0.80	2.13	2.02	1.83	1.64	1.46	1.29	1.12	0.94	0.75	0.51
0.90	3.01	2.86	2.58	2.32	2.07	1.83	1.58	1.33	1.05	0.73
0.99	9.51	9.05	8.16	7.34	6.55	5.77	5.00	4.20	3.33	2.29

Relative standard error ratio : blue - metrology advantage, green - VM advantage
Independent population number and process variation

Calculation Formula

Analyzing the statistical Standard Error of the mean value

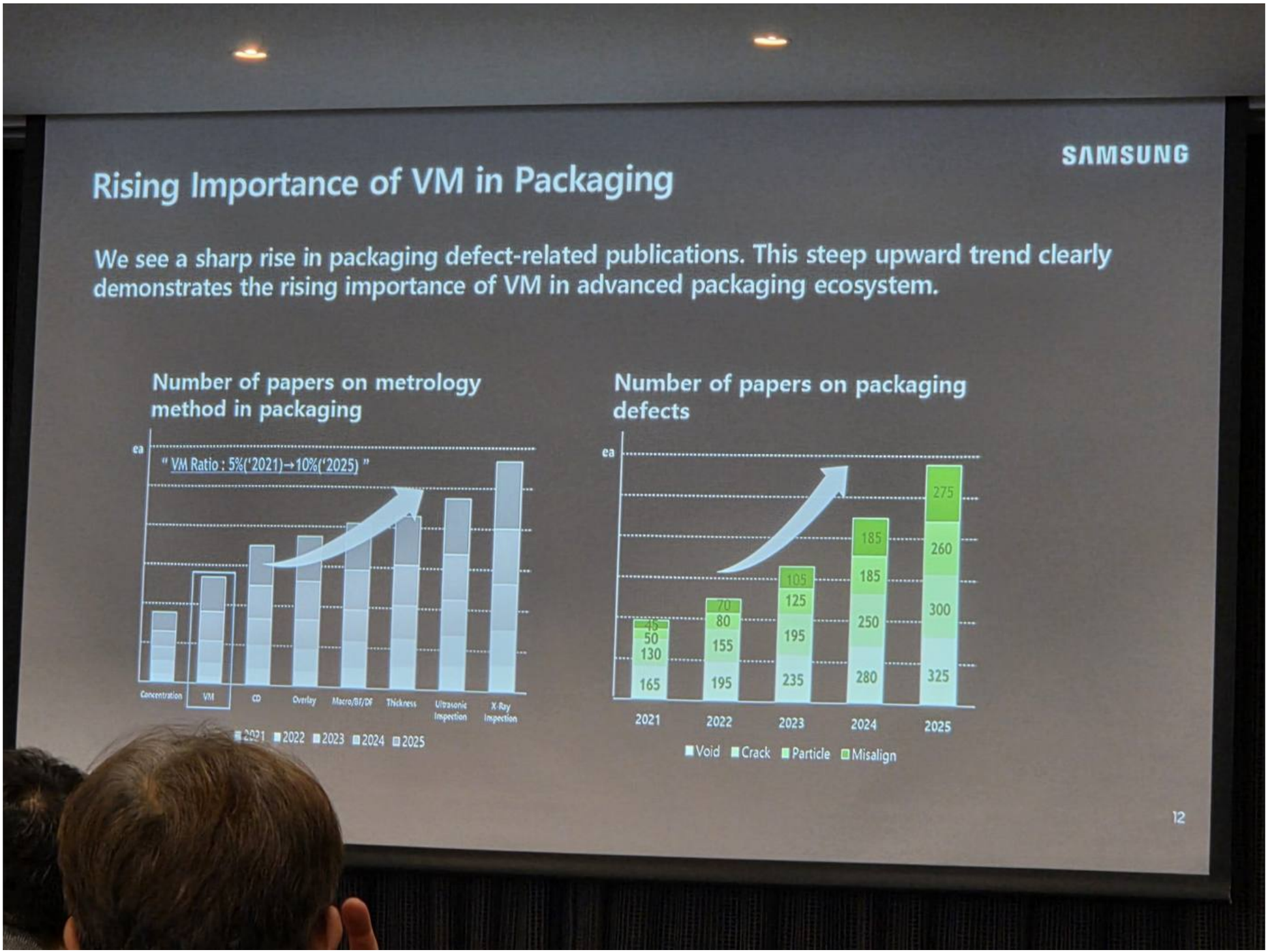
$$U_{metrology}{}^2 = U_{sampling}{}^2 \propto \frac{s^2}{n}\left(\frac{N-n}{N-1}\right)$$

$$U_{VM}{}^2 = U_{modeling}{}^2 + U_{bias}{}^2 \propto \frac{s^2(1-R^2)}{N} + \frac{s^2(1-R^2)}{n}$$

$$\therefore \left(\frac{U_{metrology}}{U_{VM}}\right)^2 = \left(\frac{1}{\alpha^2}\right)\frac{(N-n)}{(1-R^2)(N+n)}$$

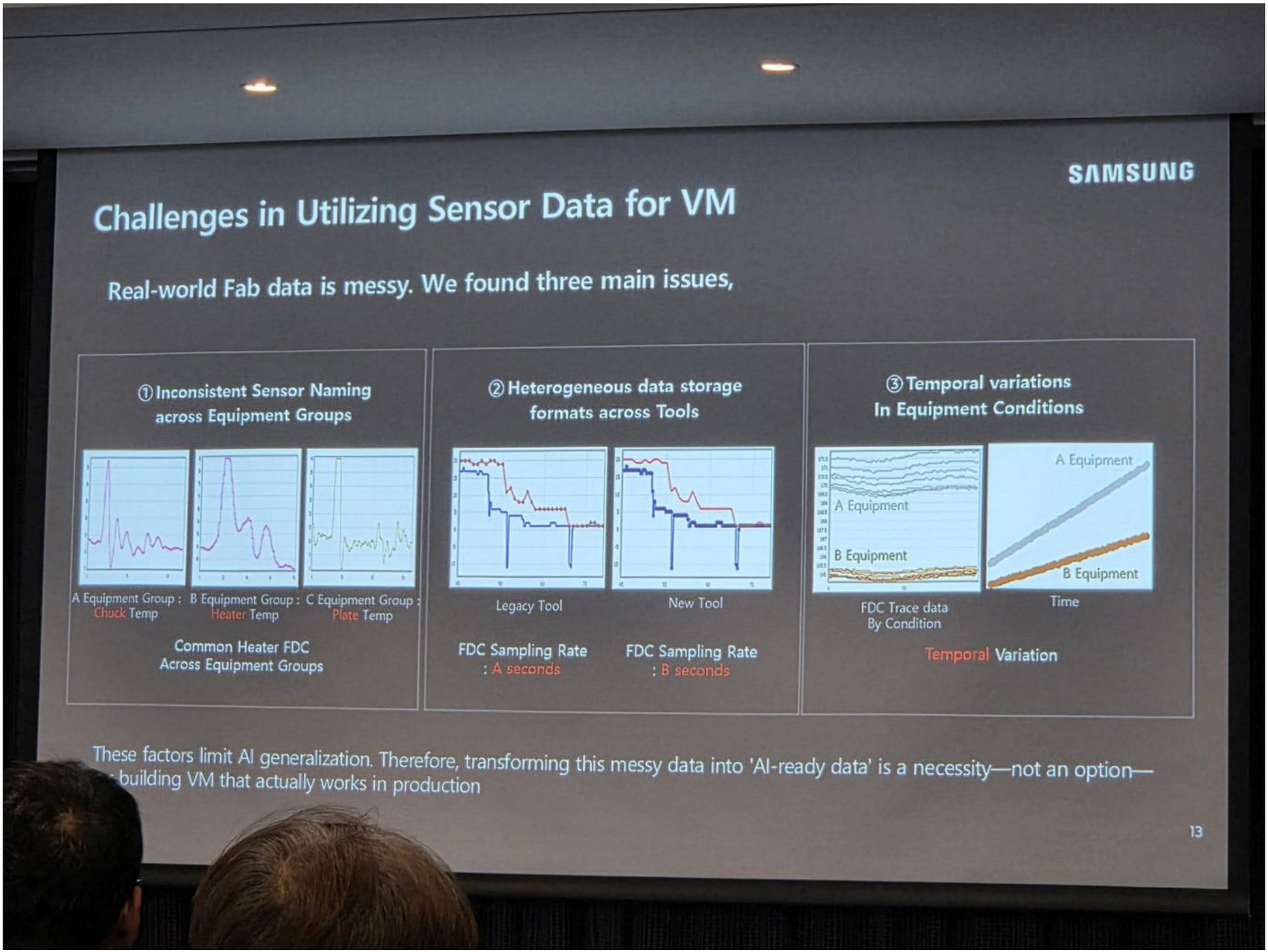
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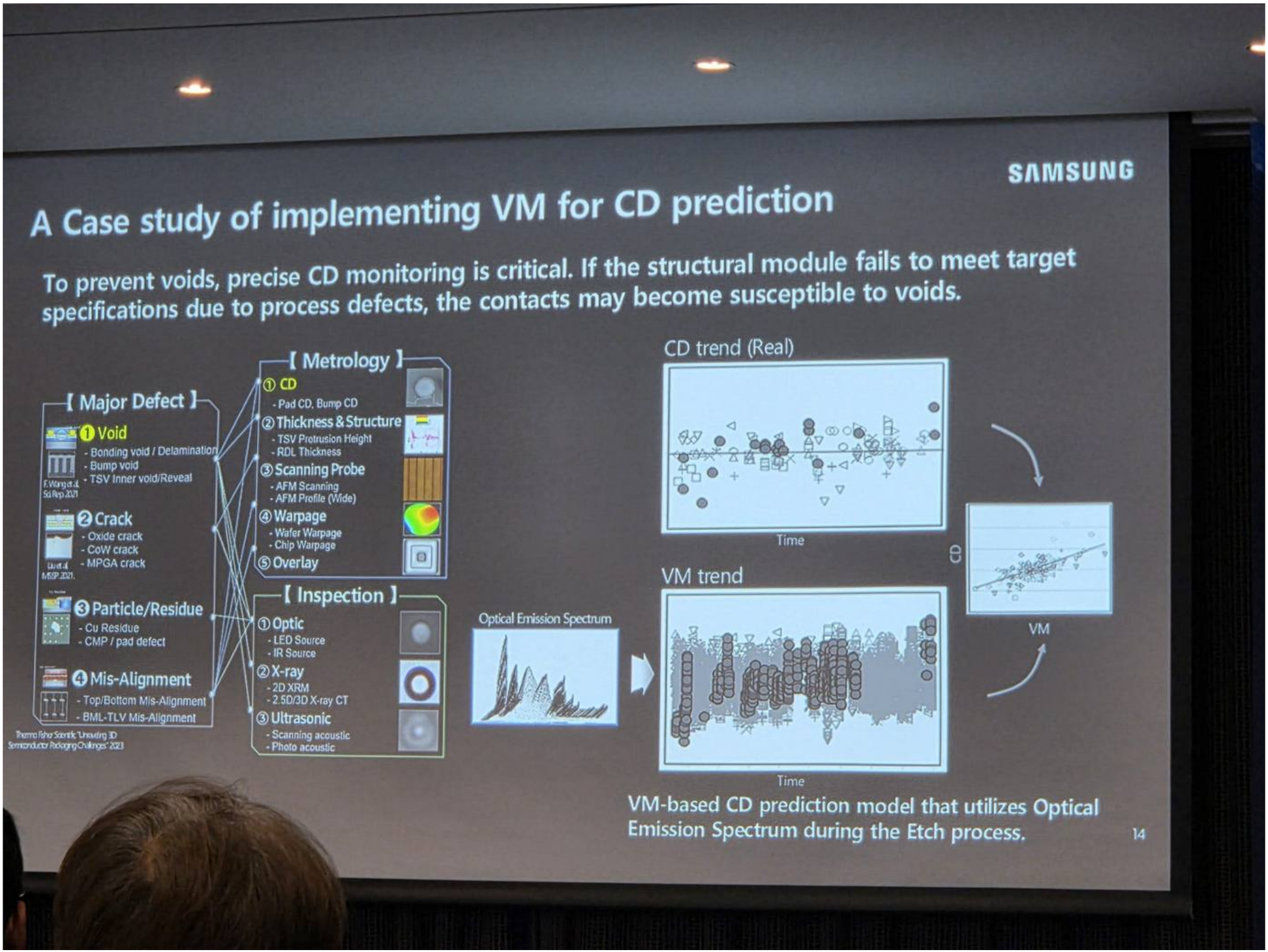


T03 첨단 패키징 가상계측과 이상관리

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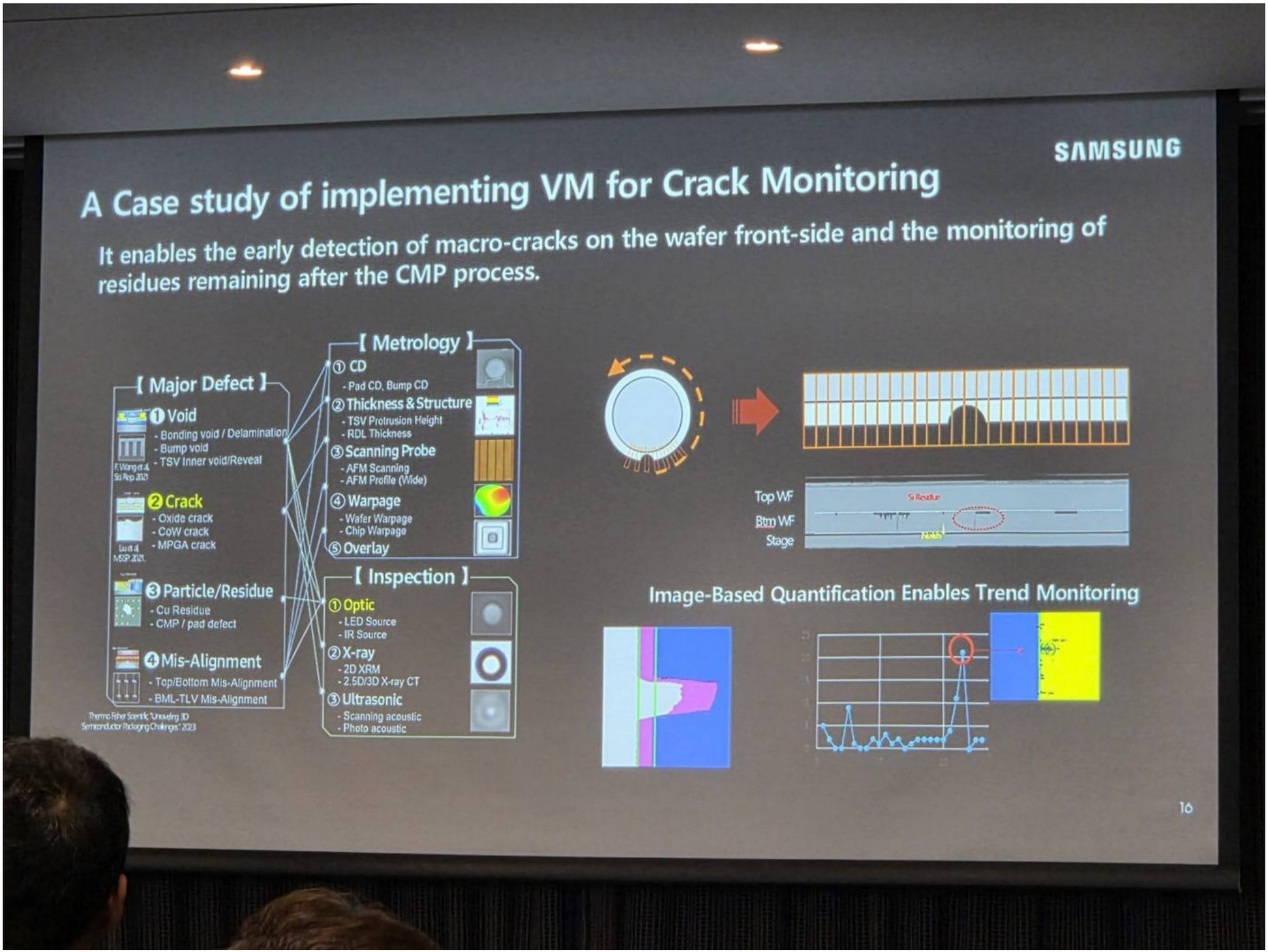


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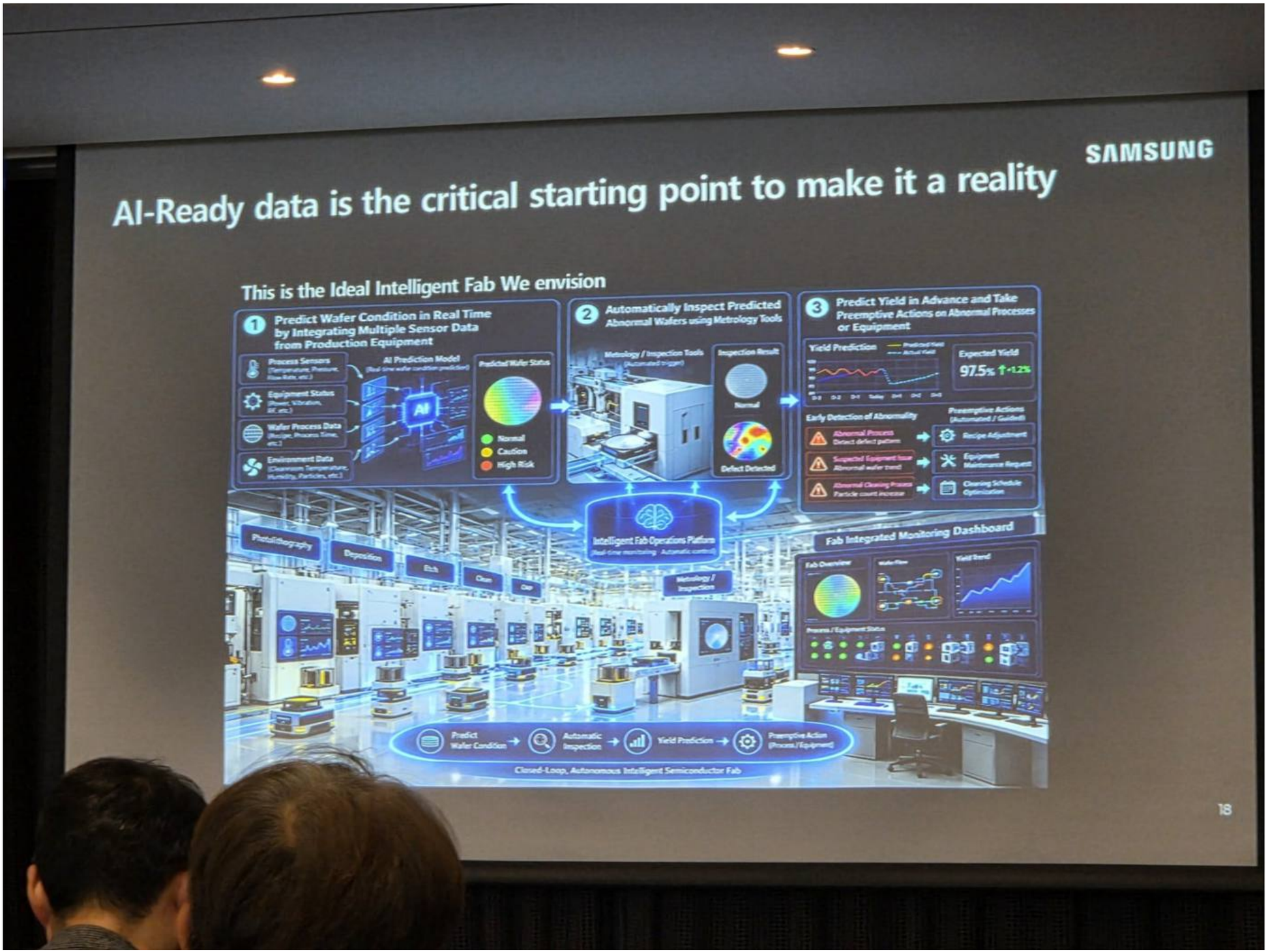


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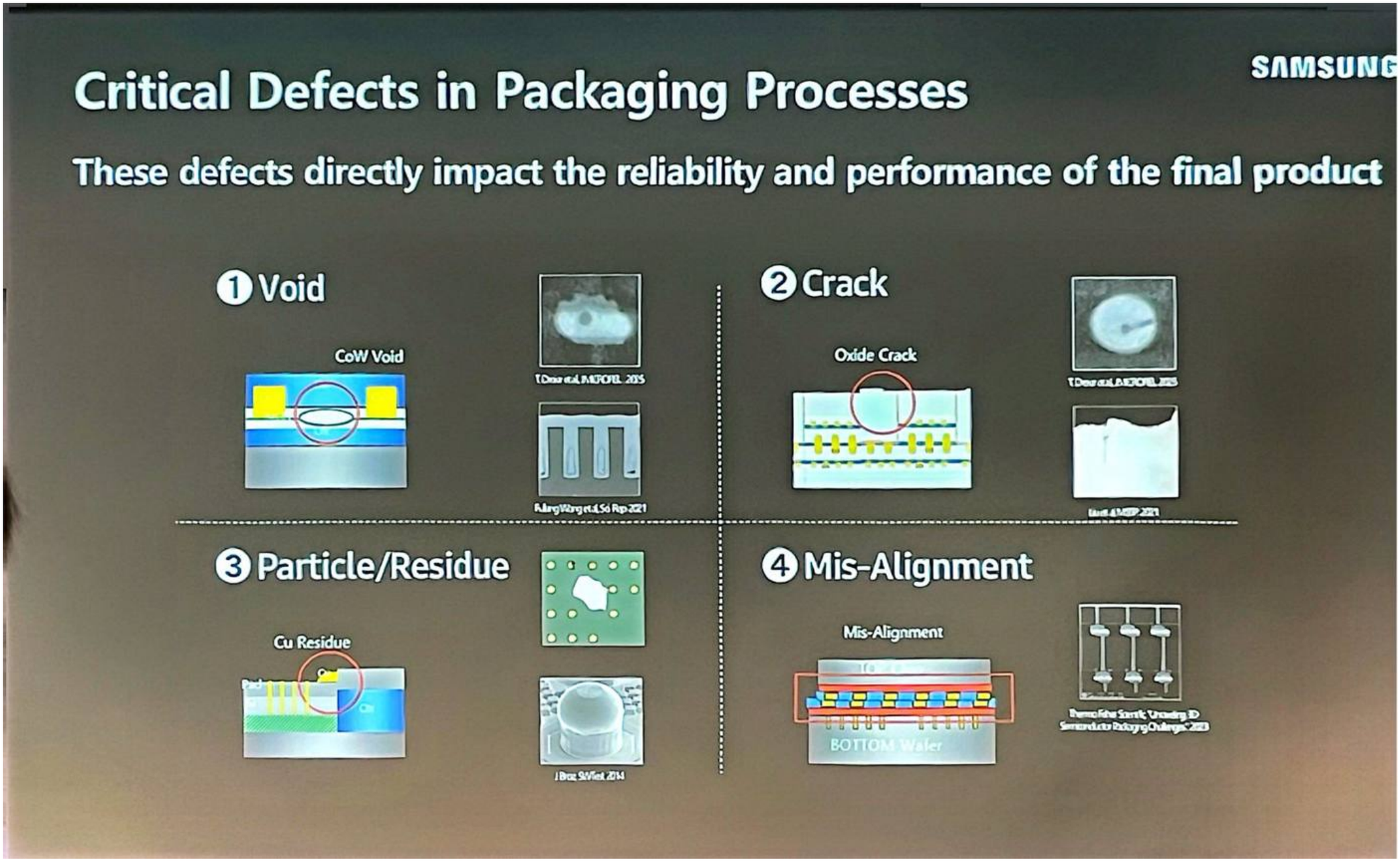


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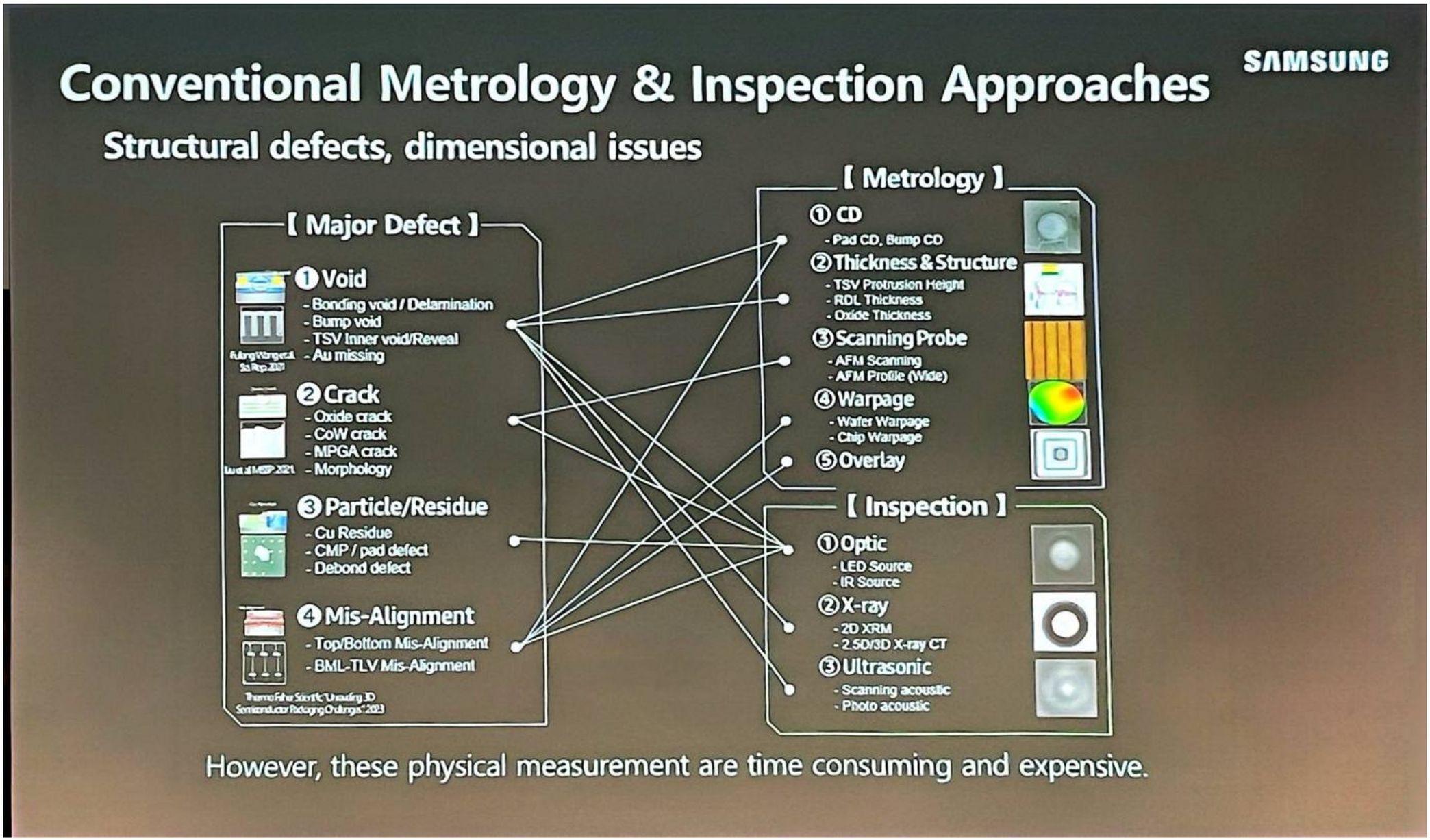
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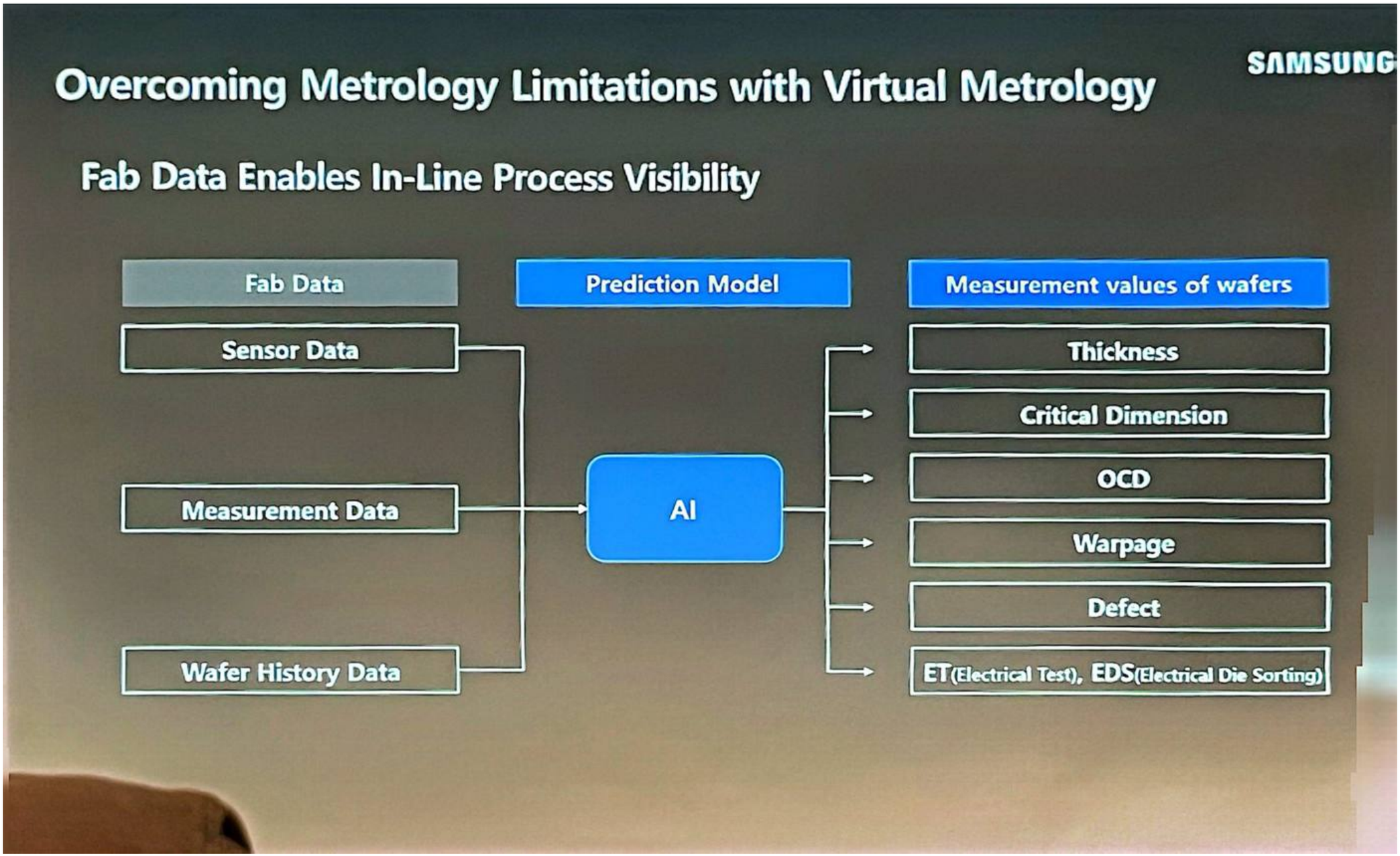
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